

This project estimates the nutritional content of Indian dishes based on user-provided ingredients and quantities. It uses a nutrition database and household measurement approximations to calculate energy, carbs, protein, fat, and fibre per standard serving (typically 200ml katori).

Features:

- Load and clean nutrition data from a CSV
- Convert various household measurements (e.g., cups, tablespoons) to grams
- Classify dish types (e.g., 'Wet Sabzi', 'Dal')
- Calculate total nutritional values
- Fuzzy match ingredients not found exactly in the database
- Output estimated nutrition per 200ml serving

Sample Output:

```
{  
  "estimated_nutrition_per_200ml_katori": {  
    "energy_kcal": 210.5,  
    "carb_g": 11.3,  
    "protein_g": 6.8,  
    "fat_g": 15.2,  
    "fibre_g": 2.1  
  },  
  "dish_type": "wet sabzi",  
  "ingredients_used": [  
    { "ingredient": "Paneer", "quantity": "0.75 cup cubes" },  
    { "ingredient": "Butter", "quantity": "2 teaspoons" },  
    { "ingredient": "Tomato", "quantity": "0.5 cup puree" },
```

```
{ "ingredient": "Onion", "quantity": "0.5 cup chopped" },  
  
{ "ingredient": "Cream", "quantity": "1 tablespoon" }  
  
]  
  
}
```

Requirements:

- Python 3.7+
- pandas
- difflib (standard Python library)

Install dependencies:

```
pip install pandas
```

Project Structure:

```
nutrition_estimator/
```

```
??? nutrition_pipeline.ipynb    # Jupyter Notebook with code blocks
```

```
??? Nutrition source.csv       # Nutrition database (must include 'food_name' column)
```

```
??? README.pdf
```

How to Use:

1. Place your Nutrition source.csv in the project folder.
2. Run the notebook cells step by step.
3. Use the process_dish(dish_name, nutrition_db) function with your dish name.
4. Output will show estimated nutrition and log any missing or unmatched ingredients.

Notes:

- Ingredient names must be as close as possible to those in the database.

- Fuzzy matching handles typos or slight name mismatches (e.g., 'onion' vs 'onions').
- Units must be in recognizable format like cup, tablespoon, teaspoon, katori.

To Do:

- Add fuzzy matching for better accuracy
- Integrate with an OCR pipeline to extract ingredients from recipe images
- Export results to JSON or Excel

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